DATA ANALYTICS PROJECT

Sales Data Analysis

1. Project Overview

This project focuses on analysing sales data to understand business performance. Sales data contains information about Customers, Products, Discount. The purpose of this project is to convert raw sales data into meaningful insights that can be easily understood.

2. Problem Statement

The company collects sales data regularly, but the data is stored in raw format. It is difficult to understand business performance by looking only at numbers. The company does not clearly know which products perform well, which regions generate low profit, and how sales change over time.

3. Project Objectives

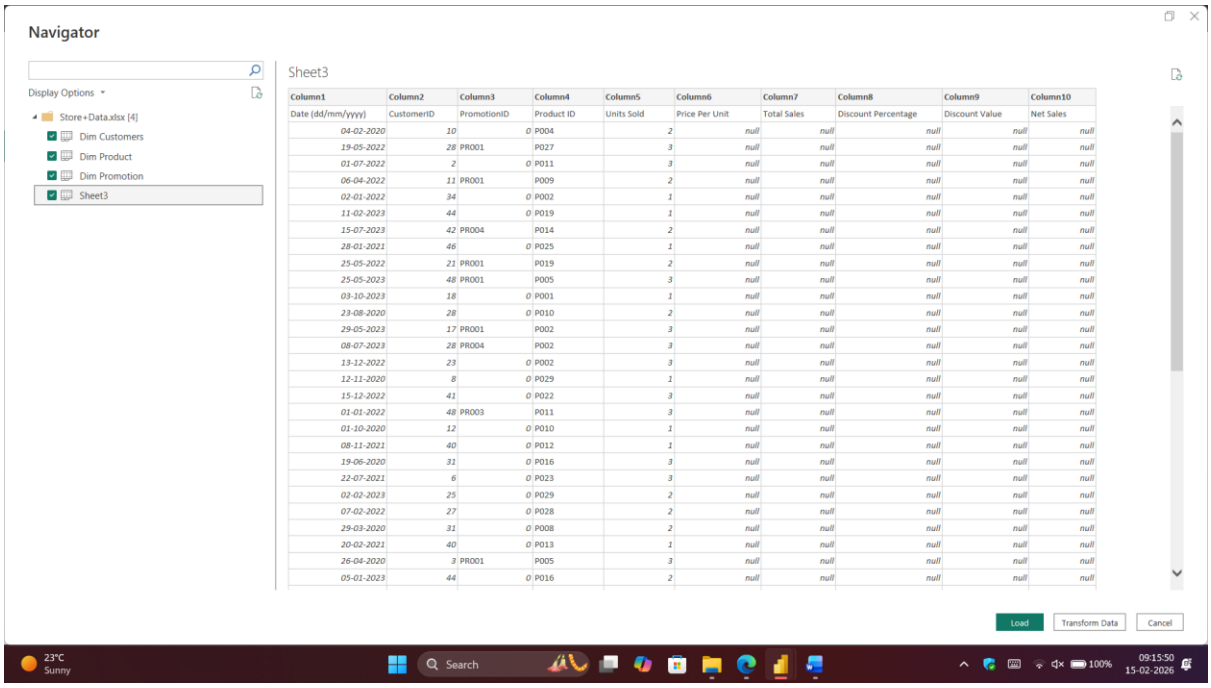
- Top/Bottom 5 product by Sales/profit/Quantity Sold.
- sales trends vary over time (daily, monthly, quarterly, annually).
- Relationship between sales & profit.
- Average discount offered in each discount category.
- Total number of orders.
- Sales by different cities.

4. Data Description

The dataset used in this project is a sales dataset stored in a CSV file. It contains five years of sales records including order date, product name, category, region, sales, and units sold.

5. Data Understanding

The dataset was reviewed to understand the number of rows and columns, data types, and missing values.

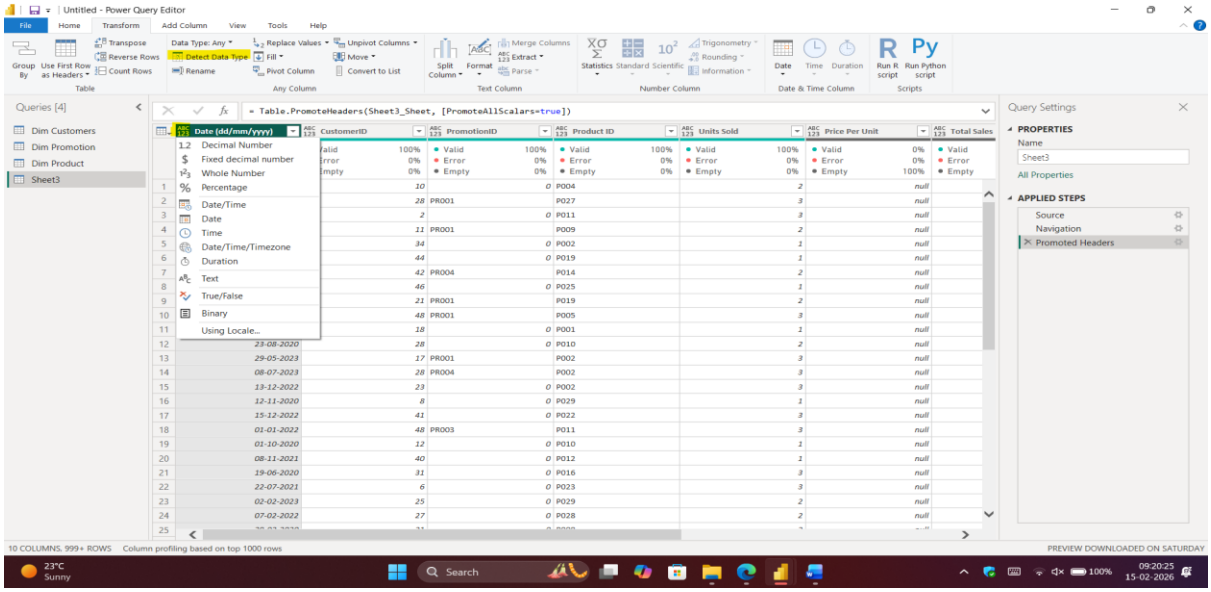


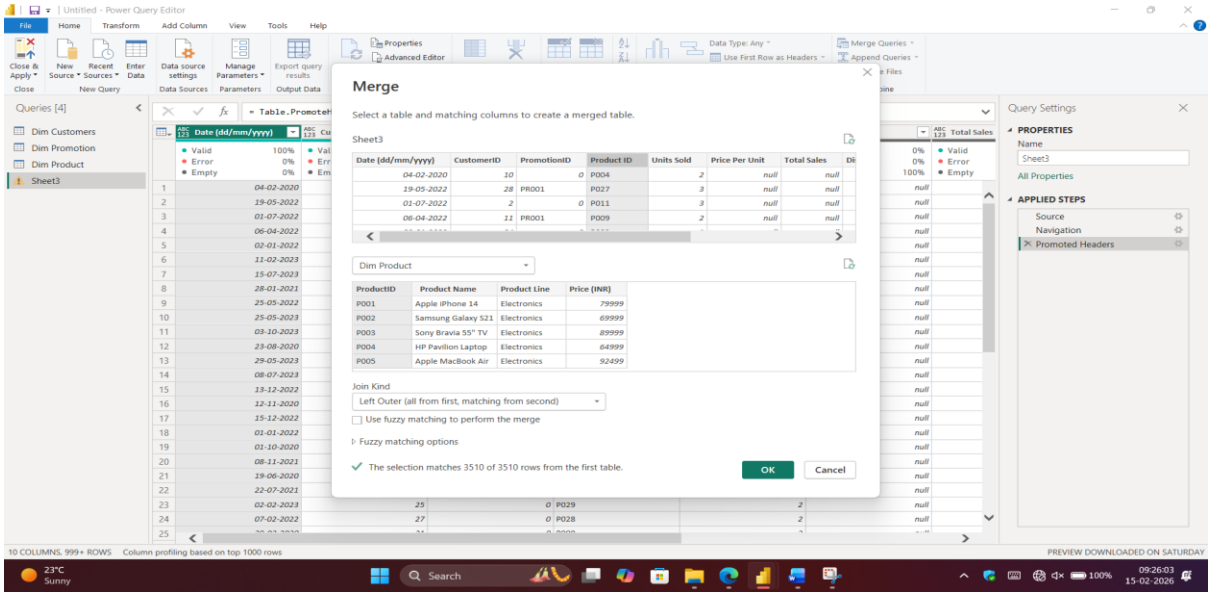
Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10
Date (dd/mm/yyyy)	CustomerID	PromotionID	Product ID	Units Sold	Price Per Unit	Total Sales	Discount Percentage	Discount Value	Net Sales
04-02-2020	10	0	P004	2	null	null	null	null	null
19-05-2022	28	PR001	P027	3	null	null	null	null	null
01-07-2022	2	0	P011	3	null	null	null	null	null
06-04-2022	11	PR001	P009	2	null	null	null	null	null
02-01-2022	34	0	P002	1	null	null	null	null	null
11-02-2023	44	0	P019	1	null	null	null	null	null
15-07-2023	42	PR004	P014	2	null	null	null	null	null
28-01-2021	46	0	P025	1	null	null	null	null	null
25-05-2022	21	PR001	P019	2	null	null	null	null	null
25-05-2023	48	PR001	P005	3	null	null	null	null	null
03-10-2023	18	0	P001	1	null	null	null	null	null
23-08-2020	28	0	P010	2	null	null	null	null	null
29-05-2023	17	PR001	P002	3	null	null	null	null	null
08-07-2023	28	PR004	P002	3	null	null	null	null	null
13-12-2022	23	0	P002	3	null	null	null	null	null
12-11-2020	8	0	P029	1	null	null	null	null	null
15-12-2022	41	0	P022	3	null	null	null	null	null
01-01-2022	48	PR003	P011	3	null	null	null	null	null
01-10-2020	12	0	P010	1	null	null	null	null	null
08-11-2021	40	0	P012	1	null	null	null	null	null
19-06-2020	31	0	P016	3	null	null	null	null	null
22-07-2021	6	0	P023	3	null	null	null	null	null
02-02-2023	25	0	P029	2	null	null	null	null	null
07-02-2022	27	0	P028	2	null	null	null	null	null
29-03-2020	31	0	P008	2	null	null	null	null	null
20-02-2021	40	0	P013	1	null	null	null	null	null
26-04-2020	3	PR001	P005	3	null	null	null	null	null
05-01-2023	44	0	P016	2	null	null	null	null	null

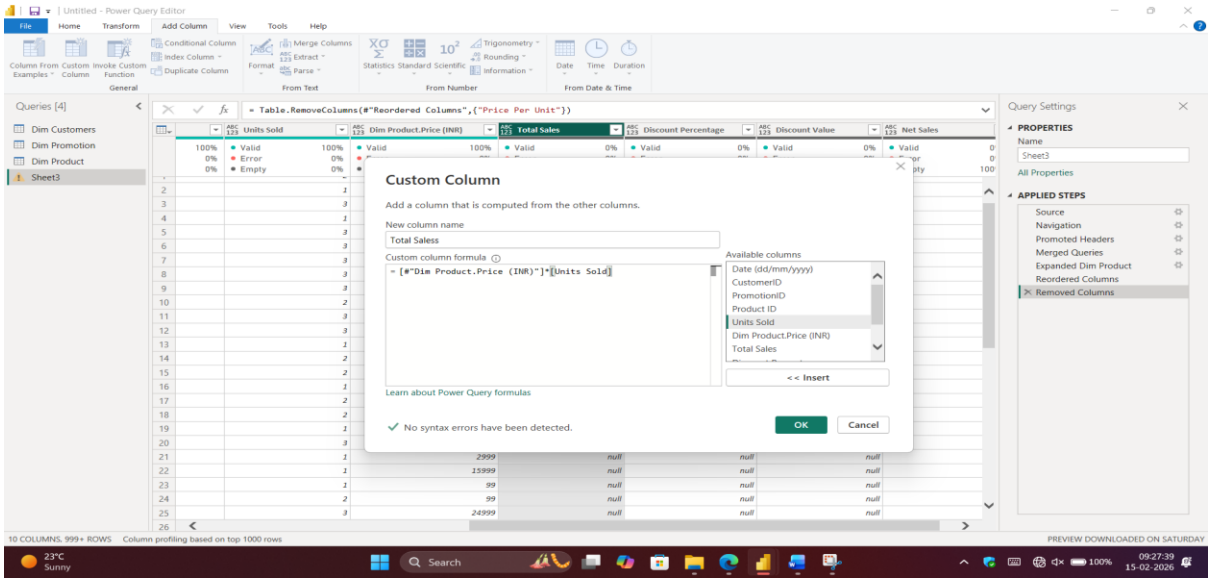
6. Data Cleaning

Duplicate records were removed, missing values were handled, and date type and numeric formats were corrected.

- Transforming data to power query editor
- Checking data types
- Removing Null values
- Merging Two Tables
- Creating New Columns


Untitled - Power Query Editor
= Table.PromoteHeaders(Sheet3_Sheet, [PromoteAllScalars=true])
Queries [4]
Dim Customers
Dim Promotion
Dim Product
Sheet3
Date (dd/mm/yyyy)
CustomerID
PromotionID
Product ID
Units Sold
Price Per Unit
Total Sales
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...
Query Settings
PROPERTIES
Name
Sheet3
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
10 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED ON SATURDAY


Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
Sheet3
Date (dd/mm/yyyy)
CustomerID
PromotionID
Product ID
Units Sold
Price Per Unit
Total Sales
Dim Product
ProductID
Product Name
Product Line
Price (INR)
P001
Apple iPhone 14
Electronics
79999
P002
Samsung Galaxy S21
Electronics
69999
P003
Sony Bravia 55" TV
Electronics
89999
P004
HP Pavilion Laptop
Electronics
64999
P005
Apple MacBook Air
Electronics
92499
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 3510 of 3510 rows from the first table.
OK
Cancel
Queries [4]
Dim Customers
Dim Promotion
Dim Product
Sheet3
Query Settings
PROPERTIES
Name
Sheet3
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
10 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED ON SATURDAY


Untitled - Power Query Editor
= Table.RemoveColumns(#"Reordered Columns",{"Price Per Unit"})
Custom Column
Add a column that is computed from the other columns.
New column name
Total Saless
Custom column formula
= [#"Dim Product.Price (INR)"]*[Units Sold]
Available columns
Date (dd/mm/yyyy)
CustomerID
PromotionID
Product ID
Units Sold
Dim Product.Price (INR)
Total Sales
<< Insert
Learn about Power Query formulas
No syntax errors have been detected.
OK
Cancel
Queries [4]
Dim Customers
Dim Promotion
Dim Product
Sheet3
Query Settings
PROPERTIES
Name
Sheet3
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
Merged Queries
Expanded Dim Product
Reordered Columns
Removed Columns
10 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED ON SATURDAY

7. Data Analysis

Analysis was performed to understand total sales, profit, category-wise performance, region-wise performance, and monthly trends.

8. Sales Analysis

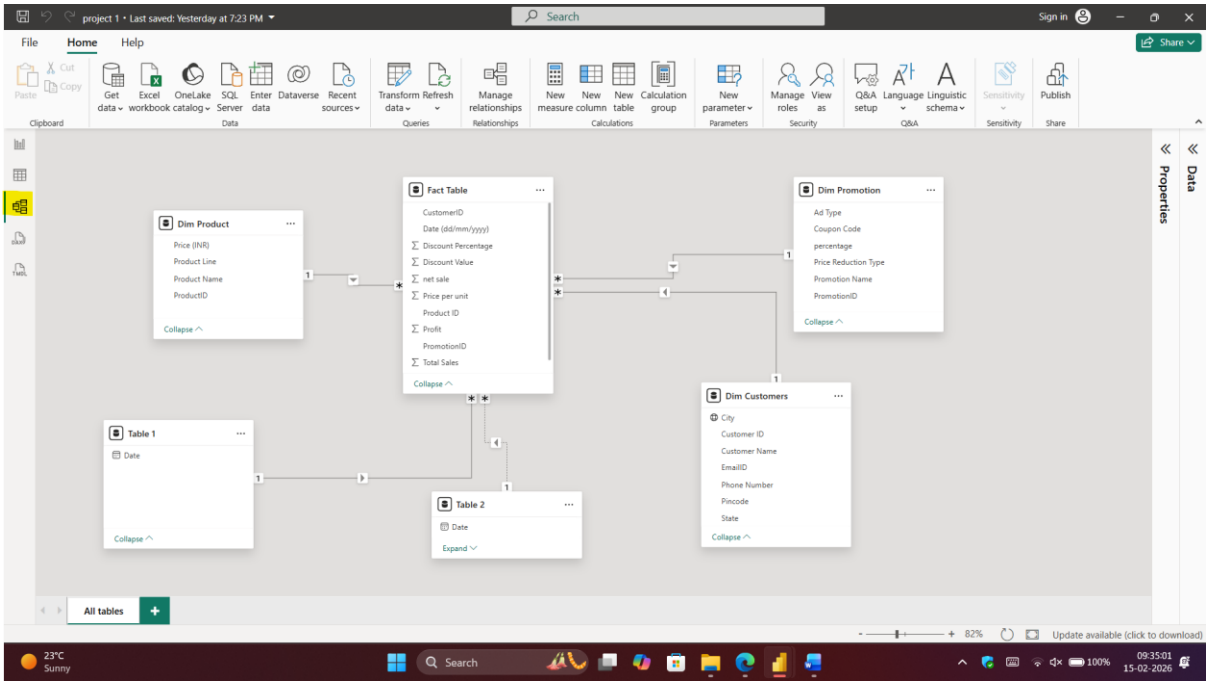
Sales analysis focused on total revenue, product-wise sales, and region-wise sales.

9. Profit Analysis

Profit analysis focused on understanding profit contribution by category and region.

10. **Relationship Between tables**

Using STAR scheme built the relationship between fact table, customer table, discount table, product table and cardinality is 1 to many (1: N).



11. Tools Used

Excel was used for understanding data and Power BI was used for cleaning, analysis and visualization.

13. Dashboard Description

The dashboard shows overall sales and profit, monthly trends, category-wise sales, and region-wise profit.

14. Key Findings

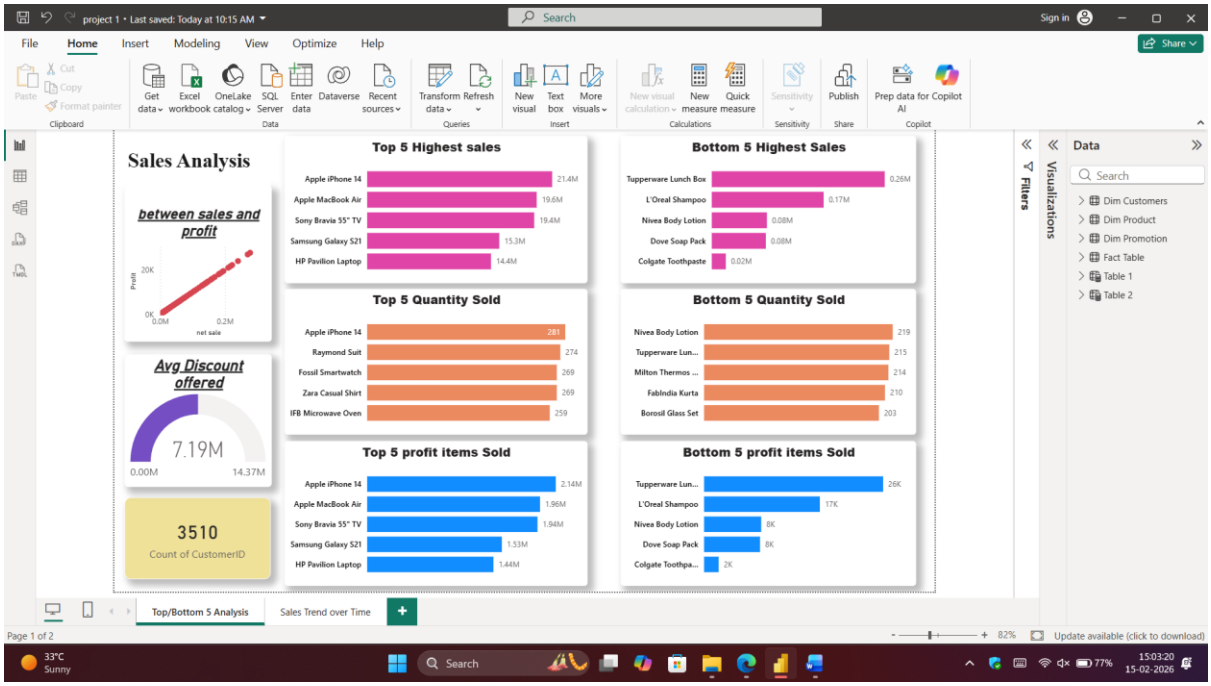
Some products contribute more to sales, some regions generate lower profit, and sales increase during certain months.

15. Conclusion

This project explains how sales data can be analyzed to understand business performance in a simple way.

16. Future Scope

The project can be extended by adding more data, customer information, and future trend analysis.


Sales Analysis
between sales and profit
Profit
net sale
Avg Discount offered
7.19M
0.00M
14.37M
3510
Count of CustomerID
Top 5 Highest sales
Apple iPhone 14 21.4M
Apple MacBook Air 19.6M
Sony Bravia 55" TV 19.4M
Samsung Galaxy S21 15.3M
HP Pavilion Laptop 14.4M
Bottom 5 Highest Sales
Tupperware Lunch Box 0.26M
L'Oreal Shampoo 0.17M
Nivea Body Lotion 0.08M
Dove Soap Pack 0.08M
Colgate Toothpaste 0.02M
Top 5 Quantity Sold
Apple iPhone 14 281
Raymond Suit 274
Fossil Smartwatch 269
Zara Casual Shirt 269
IFB Microwave Oven 259
Bottom 5 Quantity Sold
Nivea Body Lotion 219
Tupperware Lun... 215
Milton Thermos ... 214
FabIndia Kurta 210
Borosil Glass Set 203
Top 5 profit items Sold
Apple iPhone 14 2.14M
Apple MacBook Air 1.96M
Sony Bravia 55" TV 1.94M
Samsung Galaxy S21 1.53M
HP Pavilion Laptop 1.44M
Bottom 5 profit items Sold
Tupperware Lun... 26K
L'Oreal Shampoo 17K
Nivea Body Lotion 8K
Dove Soap Pack 8K
Colgate Toothpa... 2K
Top/Bottom 5 Analysis
Sales Trend over Time
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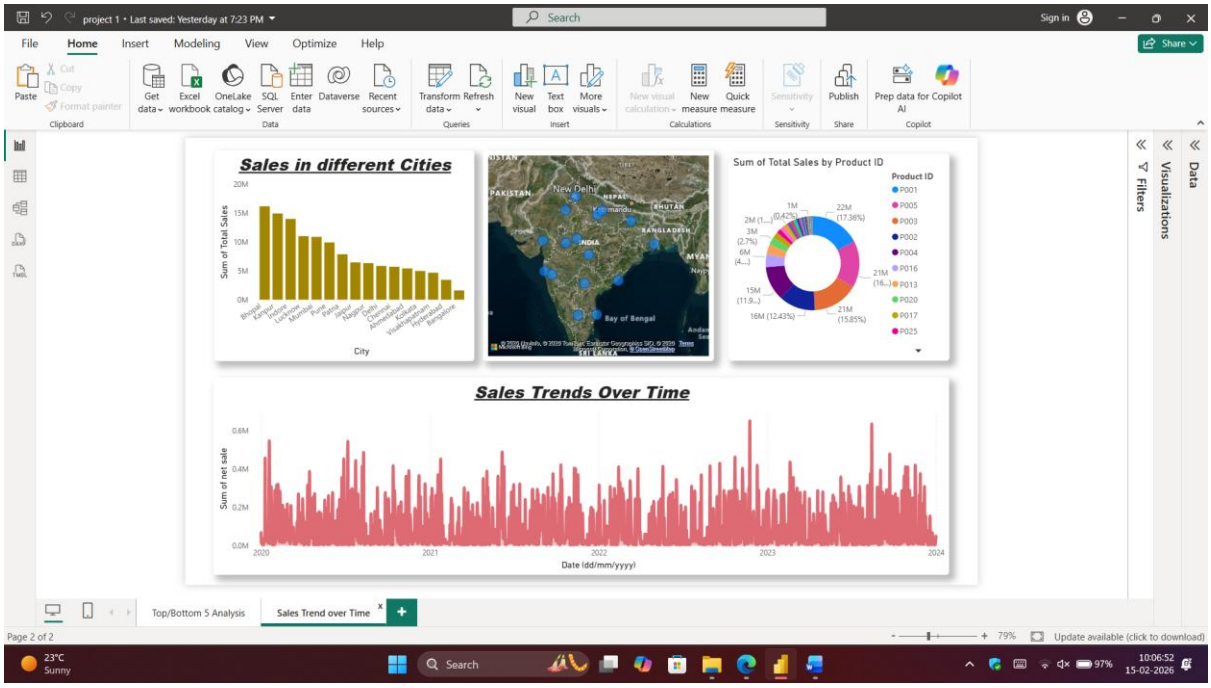
Sales in different Cities
Sum of Total Sales
City
Sum of Total Sales by Product ID
Product ID
P001
P005
P003
P002
P004
P016
P013
P020
P017
P025
Sales Trends Over Time
Sum of net sale
Date (dd/mm/yyyy)
Top/Bottom 5 Analysis
Sales Trend over Time
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